

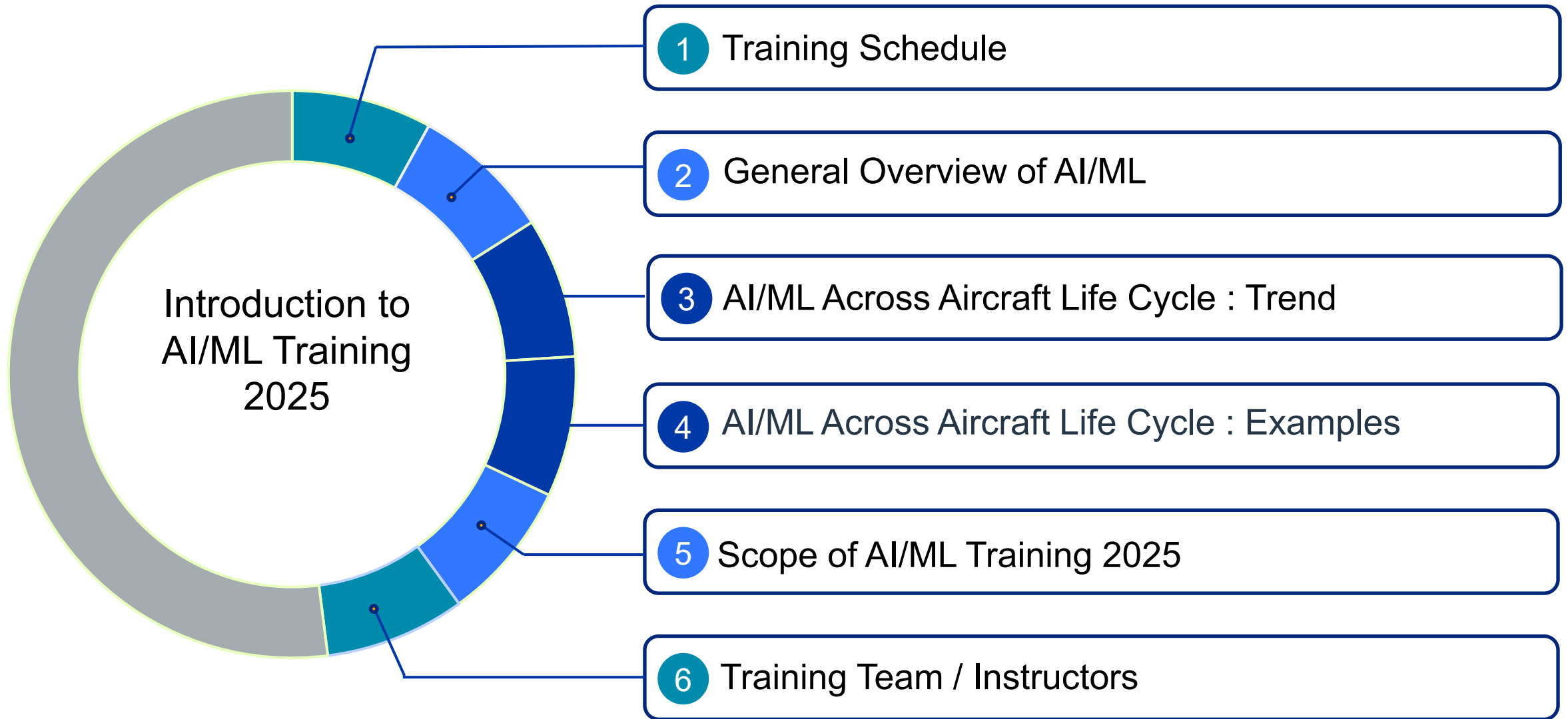


Introduction to AI/ML Training 2025

BIETC, 17th November 2025

Partha Adhikari, ATF - IVHM & Data Analytics, Boeing AI





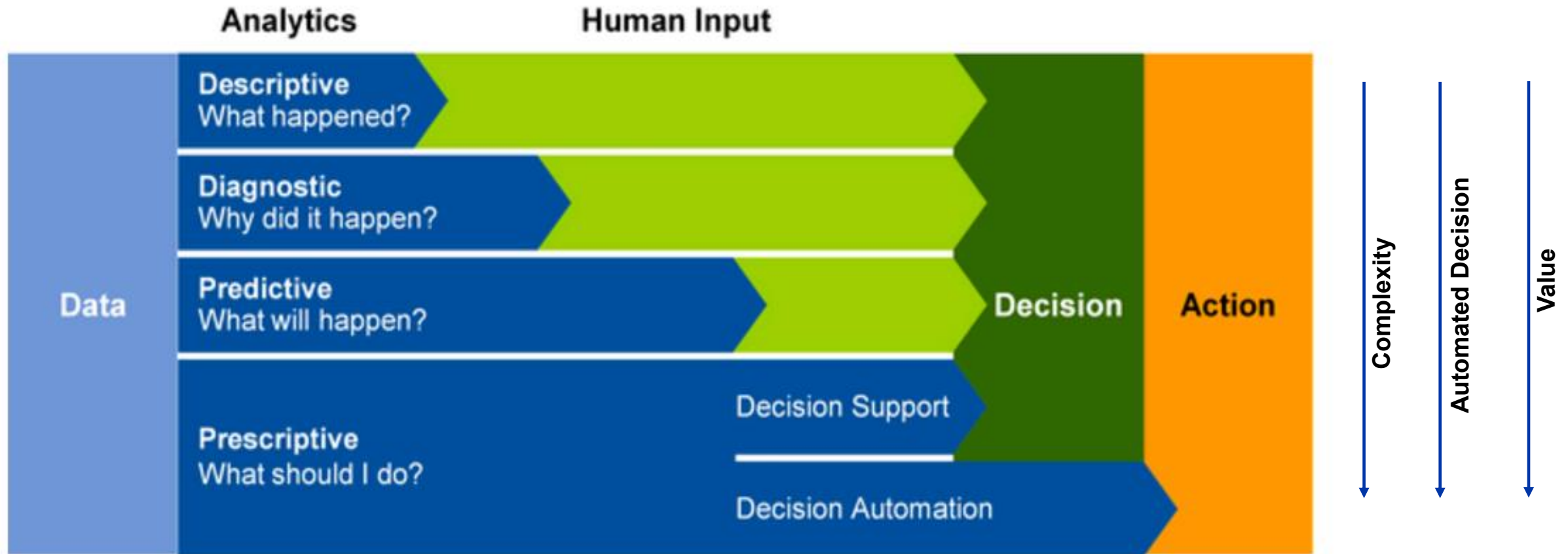
Session	Date	Day	Slot	Topic	Lecture Instructor	Tutorial Instructor
1	17/11/2025	Monday	1	Introduction to Training	Partha	-
1	17/11/2025	Monday	1	Statistics & Hypothesis Testing	Vignesh	Ashish
1	17/11/2025	Monday	2	Statistics & Hypothesis Testing	Vignesh	Ashish
1	17/11/2025	Monday	3	Statistics & Hypothesis Testing	Vignesh	Ashish
2	18/11/2025	Tuesday	1	Exploratory Data Analysis	Deepak	Anusha
2	18/11/2025	Tuesday	2	Regression	Deepak	Anusha
2	18/11/2025	Tuesday	3	EDA & Regression	Deepak	Anusha
3	19/11/2025	Wednesday	1	Classification	Siva	Aswin
3	19/11/2025	Wednesday	2	Classification	Siva	Aswin
3	19/11/2025	Wednesday	3	Unsupervised Learning	Adarsh	Karthikeyan
4	20/11/2025	Thursday	1	Unsupervised Learning	Adarsh	Karthikeyan
4	20/11/2025	Thursday	2	Neural Networks	Rajkumar	Karthikeyan
4	20/11/2025	Thursday	3	Neural Networks	Rajkumar	Karthikeyan
5	21/11/2025	Friday	1	Neural Networks	Rajkumar	Karthikeyan
5	21/11/2025	Friday	2	Cloud Deployment	Anirban	Ashutosh Jha
5	21/11/2025	Friday	3	Cloud Deployment	Anirban	Ashutosh Jha

Slot Details	
Slot 1	09:15 AM - 11:15 AM
Tea Break	11:15 AM - 11:30 AM
Slot 2	11:30 AM - 01:00 PM
Lunch Break	01:00 PM - 02:00 PM
Slot 3	02:00 PM - 04:00 PM

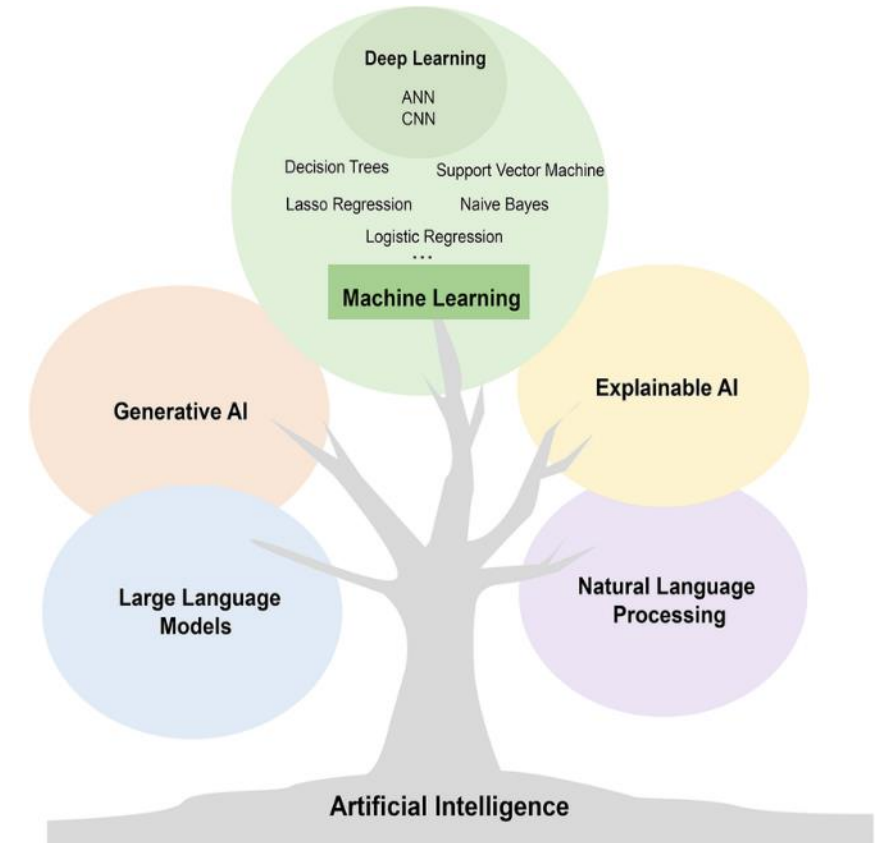
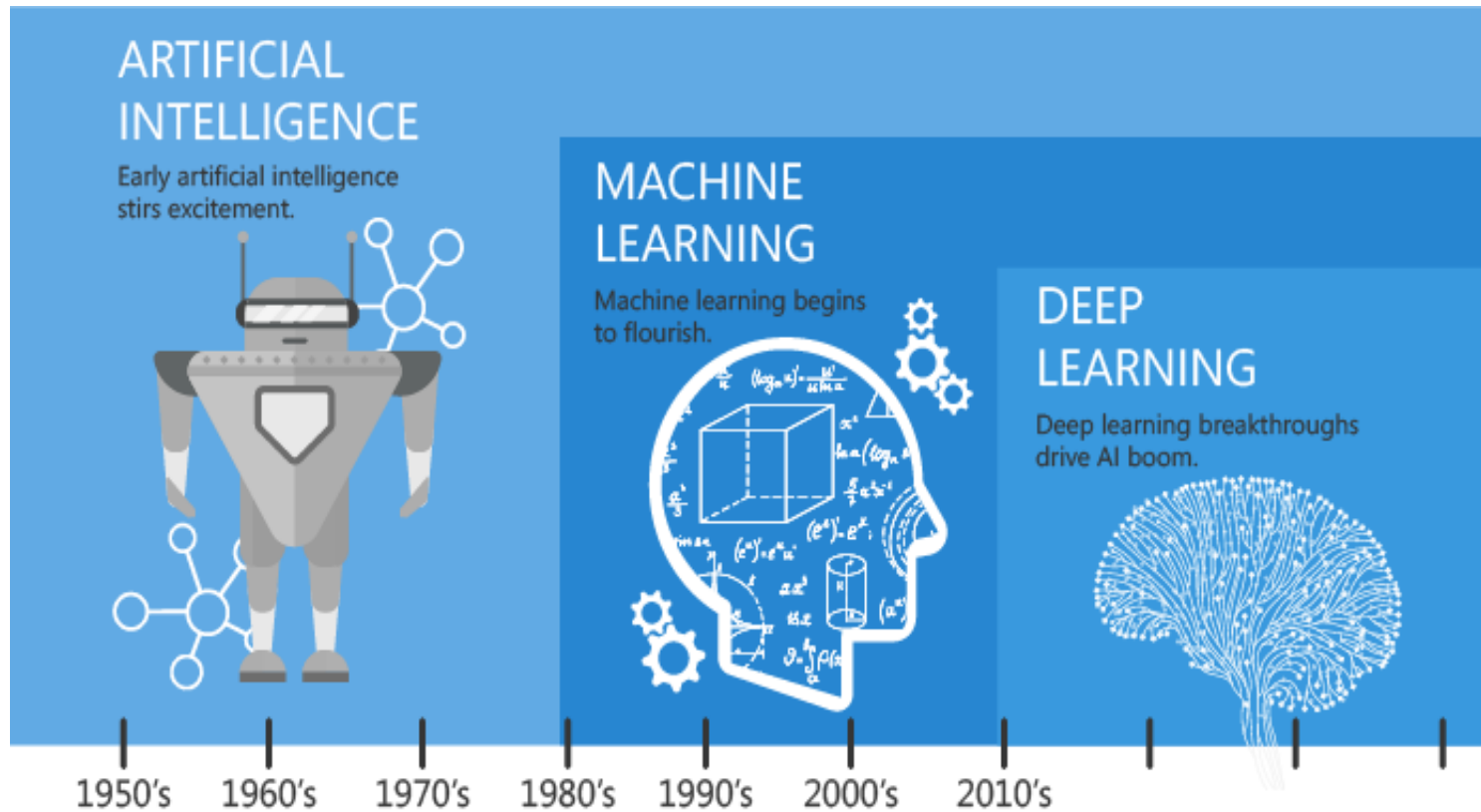
Important Dates

- Lecture Sessions: 17th -21st Nov
- Doubt Clearing Session (Virtual/Optional): 5th-Dec
- Assignment Submission Deadline: 12th-Dec

General Overview of AI/ML

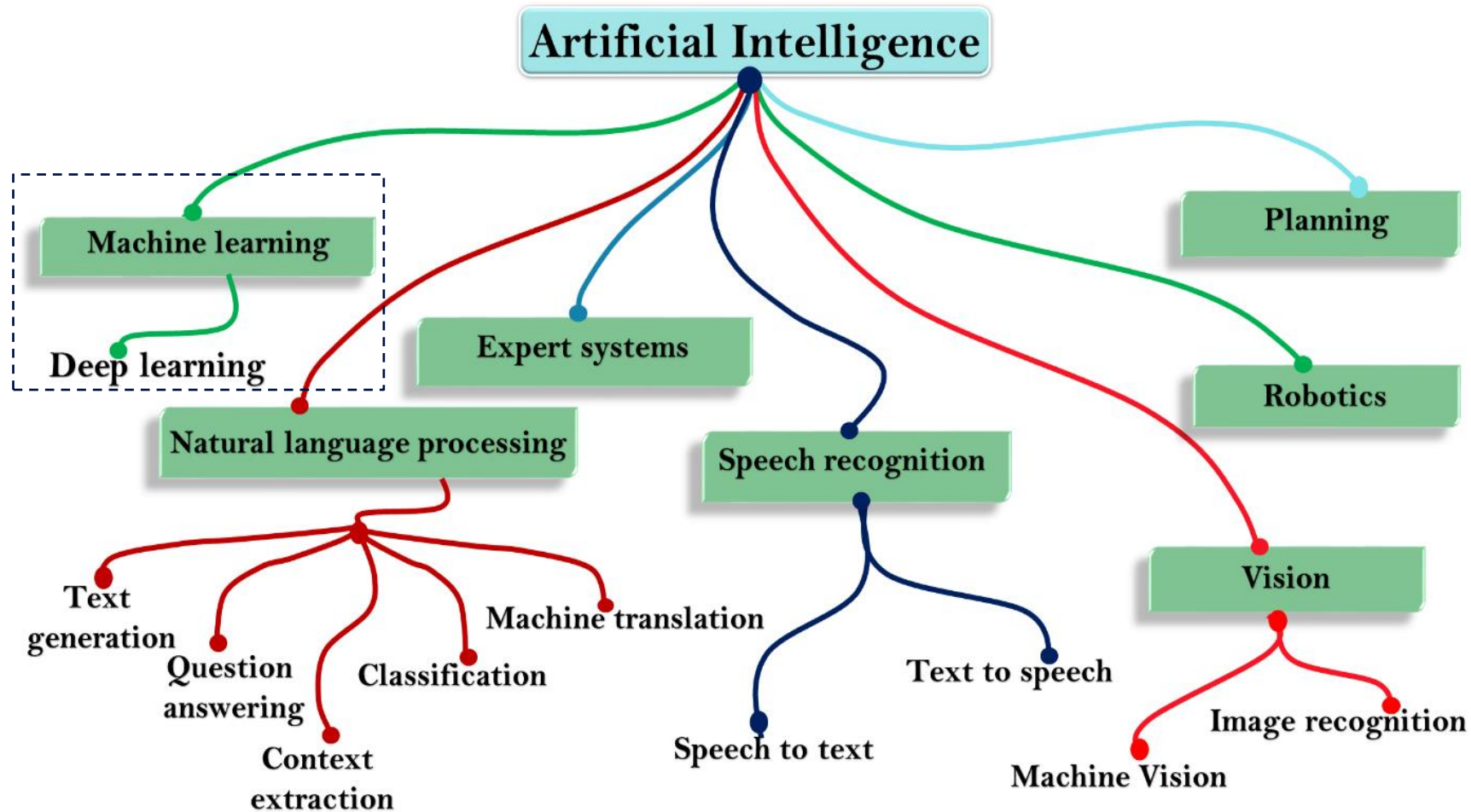


Descriptive, diagnostic, predictive, and prescriptive analytics are all phases of **data analytics**. While **data science** is the broader, overarching field that utilizes these phases, data analytics specifically focuses on the application of these four levels to extract insights and inform business decisions.



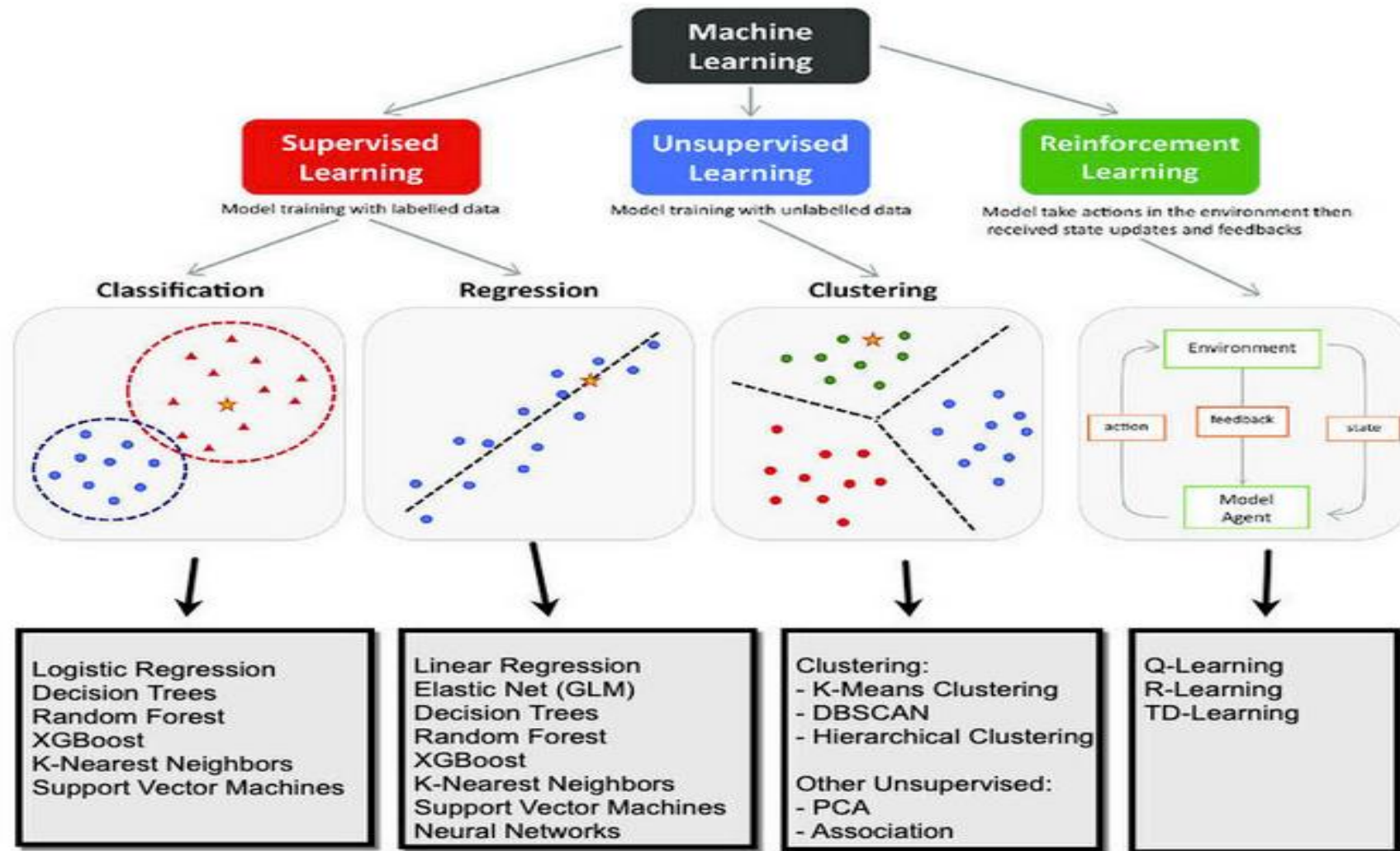
- **AI:** “The ability of machines to work and think, like the human brain, is called Artificial Intelligence (AI).”
- **ML:** “Machine learning (ML) is the tendency of machines to learn from data analysis and achieve Artificial Intelligence.”
- **DL:** “The technique used for implementing machine learning is known as deep learning (DL).”

Sources: <https://hackernoon.com/difference-between-artificial-intelligence-machine-learning-and-deep-learning-1pcv3zeg>
https://www.researchgate.net/figure/The-general-workflow-of-supervised-machine-learning-in-lymphoma-Supervised-machine_fig3_379862632



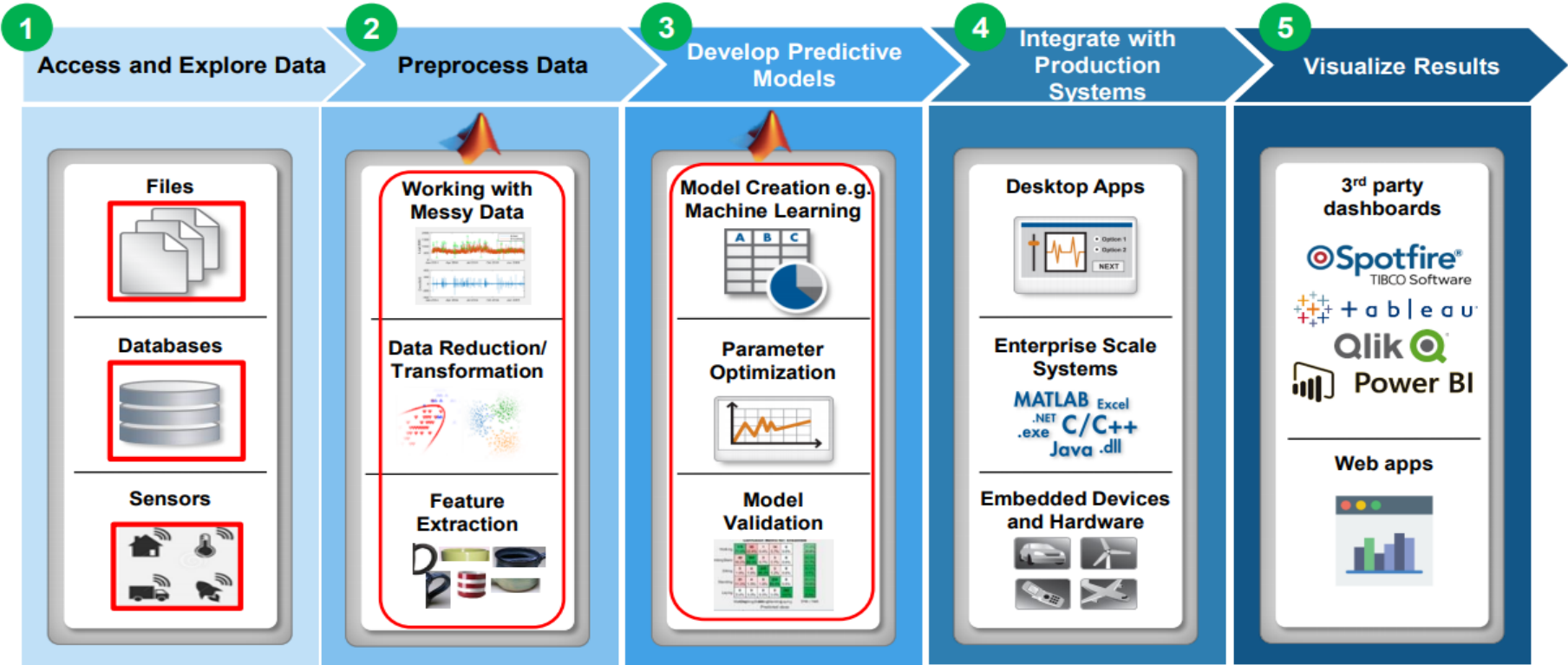
Key functions of AI : Learning, Reasoning and decision-making, Problem-solving and Perception

Source: <https://www.bntimes.com/science/real-vs-fake-ai-how-to-spot-the-difference>,
<https://www.aiacceleratorinstitute.com/what-are-the-top-7-branches-of-artificial-intelligence/>

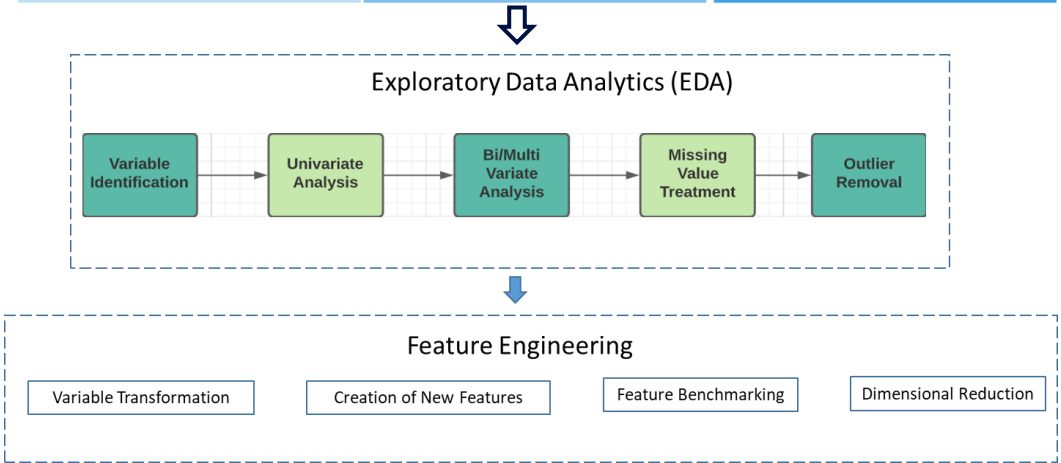


Source: <https://www.techplayon.com/machine-learning-supervised-unsupervised-reinforcement/>

- **Supervised learning:** training a model on a labeled dataset, where each training example is paired with an output label.
- **Unsupervised learning:** used when the dataset does not have labeled responses.
- **Reinforcement learning:** involves training an agent to make a sequence of decisions by rewarding it for good decisions and penalizing it for bad ones. The agent learns to achieve its goal by interacting with the environment.



Source: MathWorks



Additional References:

<https://www.analyticsvidhya.com/blog/2016/01/guide-data-exploration/>

<https://www.analyticsvidhya.com/blog/2021/03/step-by-step-process-of-feature-engineering-for-machine-learning-algorithms-in-data-science/>

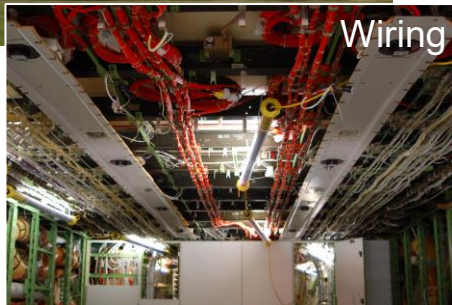
<https://www.semanticscholar.org/paper/Machine-Learning-based-Data-Driven-Diagnostics-%26-Adhikari-Rao/99d66c4f19b9123d6f73069d1a6ceef31839a2b8>

[Page 5, to find details in Feature Engineering]

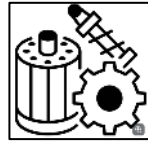
AI across Aircraft Life Cycle : Trend & Examples



787



Wiring



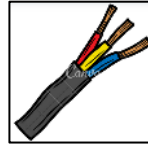
2.3 Million parts

Line Replaceable Units (LRUs)



2,00,000 sensors

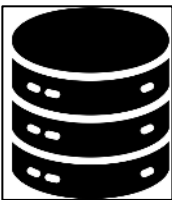
Pressure, Temperature, Strain, Acceleration, Position, etc.



70Miles cable

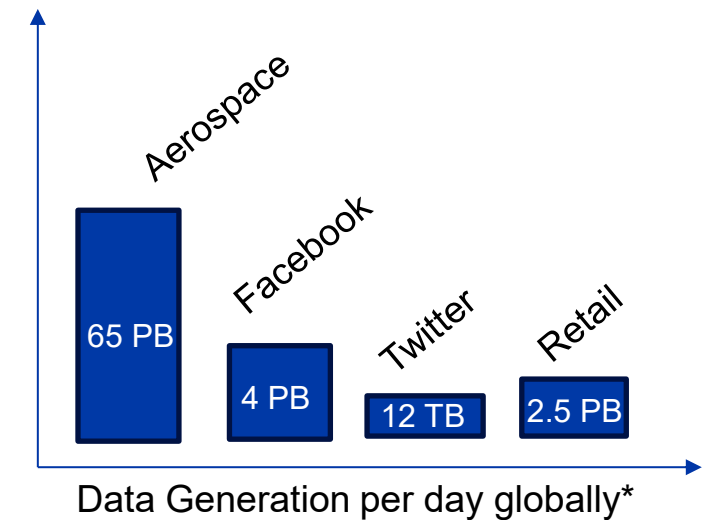


80Million lines of code

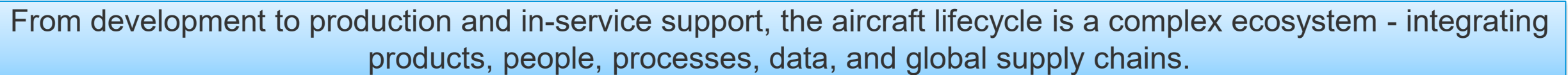


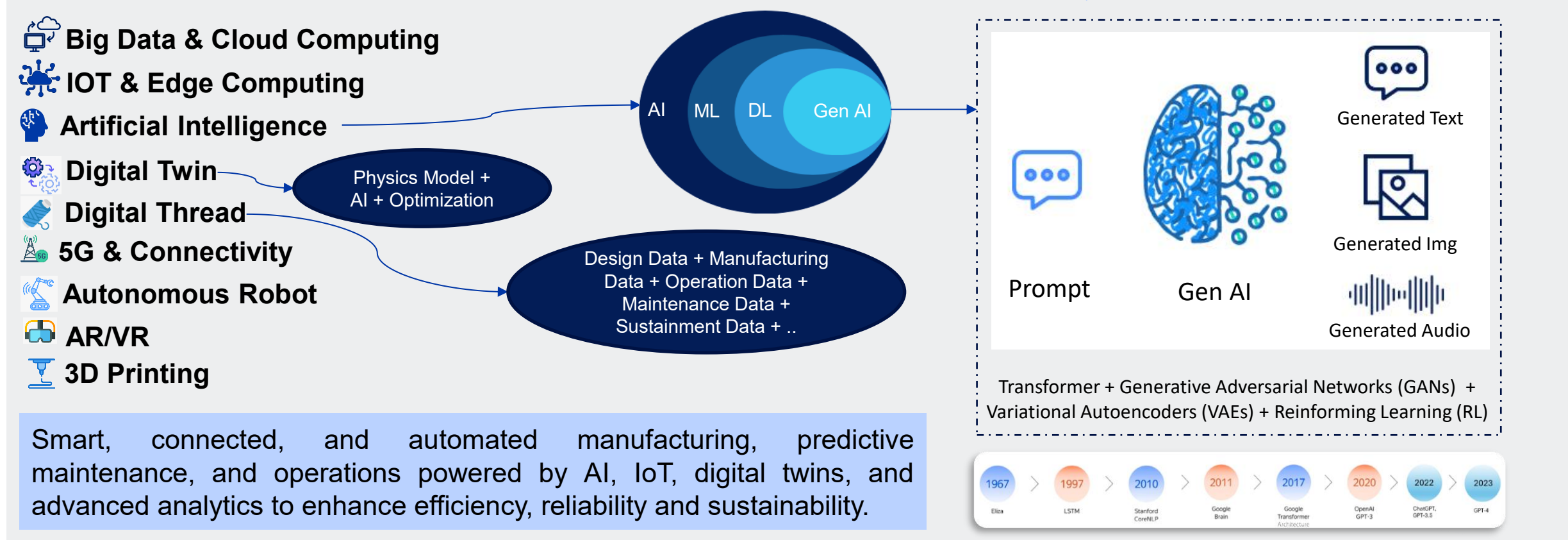
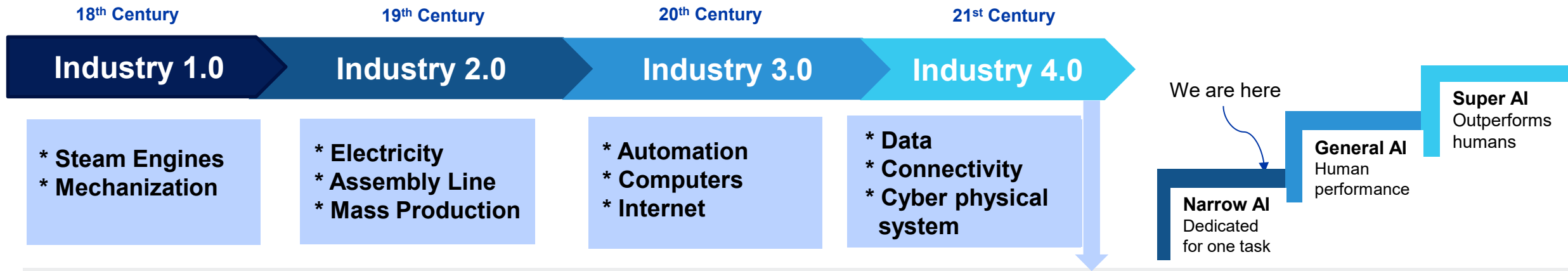
500GB+ data/flight

Avionics, Engines, Hydraulics, Flight Control, Cabin Systems, etc.

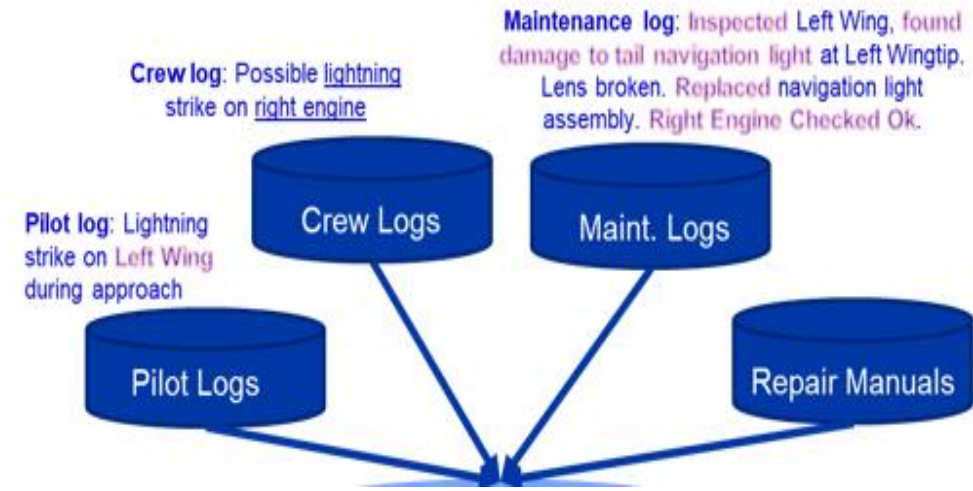


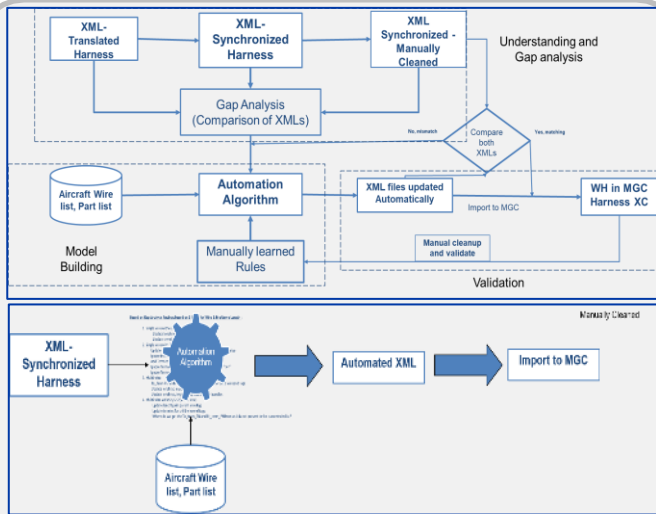
With terabytes of data generated per flight, AI-driven insights are transforming aircraft lifecycle!





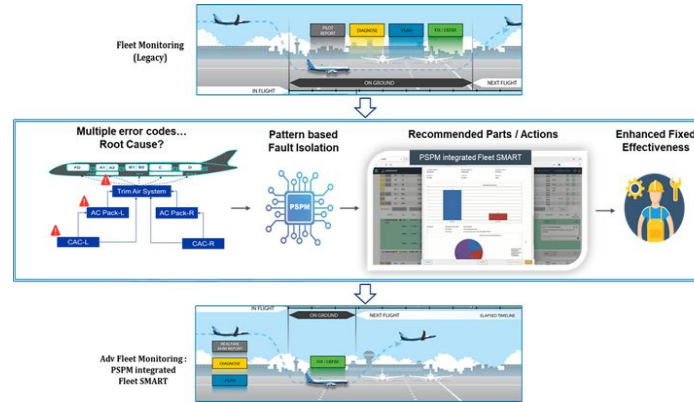
Smart, connected, and automated manufacturing, predictive maintenance, and operations powered by AI, IoT, digital twins, and advanced analytics to enhance efficiency, reliability and sustainability.





AI / ML based tool to automate wire/harness design translation

- Develop a ML based translator to automatically correct the Wiring Harness Design in Mentor Graphics (MGC) learned from CEEDS (Legacy tool) with minimal human intervention.
- BDS has realized savings of over 50,000 labor hours for ET&T India Electrical Design Integration Team.
- Customers: BDS, BIE (EDI)

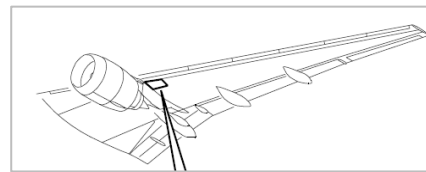


Enhanced Fault Isolation using Probabilistic Modelling

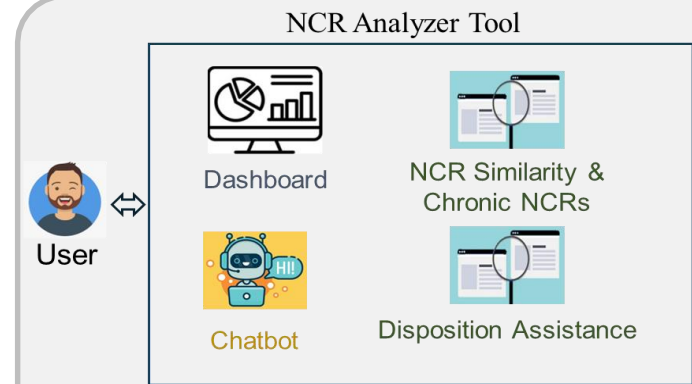
Developed solution for enhanced fault isolation & part prediction using Pattern Similarity based Probabilistic Modelling (PSPM).

Reduces incorrect part replacements by **3-6%** (cost savings of **\$265M**) during maintenances compared to existing PRITA . Transition to **BCA CS** Fleet Smart for 787, 777, 737 MAX.

Wing Anti-Ice Analytics : Failure Prognostics



- Objective was predicting wing anti-ice valve faults to reduce 777 AOGs.
- Machine Learning is used to predict valve faults, up to 12 flights prior to occurrence.
- Realized savings of **\$41M** for airlines in 2 years [**BGS**]



Non-Conformance Reports (NCR) Disposition Assistant

- Chronic NCR mapping is a labor-intensive activity requiring specialized skills
- Leverages knowledge graph, similarity-based approach, and Generative AI
- Reduces cost and labor hours spent in analyzing NCRs. started with 767, then it was extended to 737 & 777. Users: Production Engineers (ex. Liaison Engineering, Quality); customer: BCA

Boeing's principles for safe, reliable and robust AI-enabled systems define the values that guide our research, development and use of AI. These principles align with Boeing's values and are applied through our AI governance process to ensure our products contain safe and trusted systems.

Safety and Quality

- Boeing AI-enabled systems are designed, built and operated with a primary focus on safety and quality

Robustness, Reproducibility and Reliability

- Boeing AI-enabled systems behave in a safe, predictable, consistent, reliable and reproducible manner, without unexpected or surprising failure modes

Explainability, Interpretability and Transparency

- Decisions and actions of Boeing AI-enabled systems are explainable as required
- Boeing provides users clear information about the business value, training, capabilities, limitations and potential effects of Boeing AI-enabled systems

Privacy, Security and Compliance

- Boeing AI-enabled systems comply with applicable legal and regulatory requirements, specifically requirements for safety, privacy, security and performance
- Boeing AI-enabled systems have security mechanisms to mitigate the risk of unauthorized or improper use

Human Oversight, Control and Responsibility

- Boeing AI-enabled systems incorporate human oversight and control, ensuring that humans can safely override or modify system behavior during operation
- Each Boeing AI-enabled system is under the control and oversight of an accountable human as required
- Boeing AI-enabled systems execute without unintended bias, side-effects or ethical risks

- **Statistics**
 - Descriptive Statistics
 - Inferential Statistics
 - Hypothesis testing
 - Exploratory Data Analysis
 - Linear regression
- **Machine Learning**
 - Supervised Learning- Classification & Regression
 - Unsupervised learning: Clustering
 - Neural Network
 - Deep learning
- **Cloud Deployment**
 - Introduction
 - Deployment of ML algorithms in cloud

Instructors



Partha Adhikari



Rajkumar Srinivasan



Adarsh Vittal Shetty



Anirban Sarkar



Sivakumar Thiyagarajan

L&D Partners



Deepak Kumar



Vignesh Kumar



Murulidhar V



Naveen C



Aswin Mathew



Anusha Pai



Ashish Kumar Ray



Ashutosh Jha



Karthikeyan R S

Any Questions?

Thank you! 😊

